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# An effective approach for Arabic document classification using machine learning ☆,☆☆,★

Abdullah Y. Muaad [a,b,*], G. Hemantha Kumar [a], J. Hanumanthappa [a,*], J.V. Bibal Benifa [c], M. Naveen Mourya [a], Channabasava Chola [c], M. Pramodha [a], R. Bhairava [a]

[a] Department of Studies in Computer Science, University of Mysore, Manasagangothri, Mysore, 570006, India
[b] Sana'a Community College, Sana'a, 5695, Yemen
[c] Department of Computer Science and Engineering, Indian Institute of Information Technology Kottayam, India

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ABSTRACT

Arabic text classification is one application of Natural Language Processing (NLP). It has been used to analyze and categorize Arabic text. Analyzing text has become an essential part of our lives because of the increasing number of text data which makes text classification a big data problem. Arabic text classification systems become significant to maintain vital information in many domains such as education, and health sector, and public services. In the presented research work, the Arabic text classification model is developed using various algorithms namely Multinomial Naïve Bayesian (MNB), Bernoulli Naïve Bayesian (BNB), Stochastic Gradient Descent (SGD), Logistic Regression (LR), Support vector classifier (SVC), Linear SVC, and convolutional neural networks (CNN). These algorithms have been implemented utilizing the Al-Khaleej dataset. The experiments are carried out with various representation models and it is observed that CNN with character level model outperforms others. The result of CNN exceeds the state-of-the-art machine learning method with an accuracy equal to 98. The presented methods will be useful in different domains, particularly on social media.

## 1. Introduction

The problem of text mining has gained increasing attention and it became a vital research domain because of the boom of textual applications such as news article portals and social networking gates. Text classification is one of the big challenges due to the increasing number of documents. Most of the work in text classification has been done in English. Unfortunately, the number of researchers using the Arabic dataset is still very rare and needs more investigation. There are several text classification methods such as naïve bayesian, support vector machine, K-nearest neighbor (K-NN), and decision tree. Text classification refers to a data processing technique that routinely assigns a given document to its class or category. There have been numerous works in literature that deals with the subject of the classification of texts in various languages like English [1], Chinese [2], Russian [3], and many other languages. Almost more than 447 million people speak Arabic as a native speaker at the same time more than this number non-Arab but they need the Arabic language as a religious language. In addition, it is the official language of 22 countries. Moreover, many other languages consider Arabic as root. There are 29 letters/characters in the Arabic language which have to write from right to left. The Arabic language is one of the extensively spoken languages on the planet. This is the fifth most popular language in the world [4,5]. Arabic possesses a rich morphology and a complex orthography [6] which produces different words with different meanings [7,8]. This language has specific letters mentioned as Arab vowels (WAW, YAA, ALEF) (و-ي-ا)which necessitate a special system of morphology and grammar. The remaining are called consonant letters [9,10].

One of the big issues in Arabic text is handling Hamza ء (which is taken under consideration as a special case). In Arabic, two genders exist feminine and masculine. Numerical numbers are singular, double, or plural numbers. Grammatical has 3 cases: nominative, accusative, and genitive)) [11]. A noun has three linguistic cases: nominative when it is subject; accusative when it is the thing of a verb as well, genitive when it is subject to a preposition [6,12].

Classification algorithms were developed to perform text classification based on understanding the meaning which is called Classifier [13–15]. There are five steps for Arabic text such as pre-processing,

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* Corresponding author.
*E-mail addresses:* abdullahmuaad9@gmail.com (A.Y. Muaad), hanumsbe@gmail.com (J. Hanumanthappa).

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representation, dimensionality reduction, classification, and evaluation. Most of the work which has been done for Arabic text has been done using traditional ML algorithms so we are here motivated to do a design model to classify Arabic text utilizing both ML &DL approaches in the first work then extend our work to apply transfer learning in the future work. Simultaneously we have been used different type of representation to increase accuracy of text classification.

Deep learning has recently demonstrated amazing advancements for text classification as a focus area for the Arabic language. Simultaneously, a study of deep learning based on Arabic text classification has been conducted, thus we are motivated to develop the classification system for the Arabic language. The experiment has been done using a dataset called Khaleej. The practical work has been done using a different type of representation with different a classification algorithm.

The contributions of this paper are as follows,

- Proposed Arabic text classification system based on six ML algorithms and one DL.
- The experimentation part has been done using two features extraction techniques and character-level representation.
- Comparative between machine learning and deep learning with different feature extraction techniques have been done.

The proposed study investigates the best deep learning classifier for Arabic texts in terms of increasing the surfing life on the social media networks

This paper is organized as we mention here, in Section 2 related works have been explained, and the proposed model in Section 3. After that experimental analysis and results have been discuses in Sections 4 and 5 respectively. Finally, the conclusion summarizes this work in Section 6 followed by references.

## 2. Related work

Data pre-processing, feature extraction, data processing, representation, and results display are common processes in research on text analysis for Arabic texts. As we know the number of researchers in Arabic is still very less compared to English. Here are some works, we will introduce some of them in this part for example in the author in [1] explore a comparative survey of using different Arabic text classification such as traditional ML (NB) and Artificial Neural Networks. The dataset was very small equal to 4000 documents. The authors in [3] proposed text classification using N-gram frequency feature representation for language like Chinese. They design a new representation using unigram and bi-gram which give efficient results to represent and classify text. Their results prove that normalizing affected the result in text classification. The authors in [5] designed the Arabic text method to classify text from a large dataset. They used CNNs model and compare with traditional ML. The authors in [6] explore the methodology for the classification of Arab documents. The result for the classification in accuracy metrics is equal to 74.41%. The authors in [7], proposed a text classification method using the NB algorithm to classify Arabic text. The authors in [10] proposed classification methods to cluster news and classify them. The authors in [12] explore the effect of normalization with stop-word removal on text. The authors in [13] presented text classification using different representations and features extraction such as Boolean, Term Frequency (TF), Term Frequency Inverse Document Frequency (TFiDF), and others. Meanwhile, Tokenization, stop word removal, and chi-square as a selection of features was used. Lastly, the naive Bayes (NB), the decision tree (DT) and the support vector machine (SVM) has been used for classify Arabic text. The best was SVM classifier which has got accuracy equal to 95.9%. The authors in [14] compare of six different feature selection techniques with SVMs in Arabic text. The authors in [16] investigated the impact of the suppression of stop words on the categorization of Arabic documents using different algorithms and concluded that SVM with minimal sequential optimization. The authors in [17] presented a new stemmer stemming Arabic text. They conclude

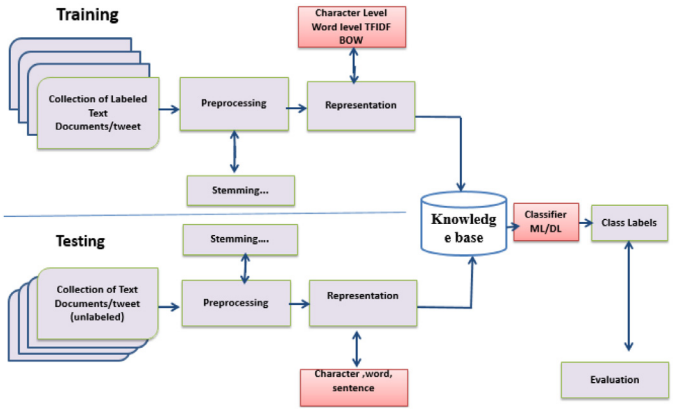



**Fig. 1.** Arabic text classification architecture.

that the Khoja stemmer was the best comparing to another stem. The authors in [18] have developed a hybrid system to categorize Arab documents based on the semantics level. Techniques such as Normalization, stop word removal, and stemming were used as pre-processing tasks. They used hybridization of N-gram + TFiDF statistical measurements to calculate similarity among words. The majority of the authors used an internally built dataset in Arabic with different sizes and contents [19]. Some researchers have worked on classifying Arab text using a different algorithm like Naïve Bayes [7,20] Support vector machine [18,21,22] Decision Trees [23] but a few of them has been used new techniques such as transfer learning. Finally, the representation of Arabic text has been used as one step for the classification task. Muaad et al. have been used character level with CNN model proposed in [24–29].

## 3. Proposed model

The paper has been implemented using machine learning and a deep learning approach. The architecture of the proposed model has been shown in Fig. 1 . There are five steps to identify and classify Arabic text as follows:

### 3.1. Pre-processing

Pre-processing is the first step in text classification and it is mainly referred to pre-paration of text data to be ready for further processing. At the same time, it refers to the process of preparing text to be a suitable form for the representation step. In this experimentation pre-processing phase contain different methods such as eliminating non-Arabic words, stopwords removal, and elimination of punctuations from input data. There are various techniques for pre-processing consisting of, (a). Tokenization, (b). Normalization, (c). Stop Word Elimination, and (d) Stemming.

### 3.2. Representation and feature engineering

Representation of text and choosing suitable futures are the most important steps to make data ready for further processing so that machines can understand. In this paper, we have used different representation models based on word and character levels at the same time different feature engineering techniques

### 3.3. Feature extraction

In this work two feature extraction techniques have been used called BOW and TFIDF. At the same time, we have been used character level to represent text with deep learning utilizing CNN. Three different methods have been used as follows:

**Table 1**
Dataset distribution.

| Category name | Number of documents |
|---|---|
| Culture | 6500 |
| Finance | 6500 |
| Medical | 6500 |
| Politics | 6500 |
| Religion | 6500 |
| Sports | 6500 |
| Tech | 6500 |
| Total | 45500 |

- Bag of words (BOW)
- TF-IDF
- Character level

### 3.4. Dimensionality reduction

Due to sparse matrices after representing text using the traditional technique so the decreasing number of features became compulsory. This number is too large for that reason we need to use dimensionality reduction techniques. After that, data is being split as training set and testing set. Then passed to classification algorithm which builds a model. Later.

### 3.5. Classification

Classification of text and classifying it to true labeled based on its content is known as classification. Several algorithms have been used as we will explain in part 5.

In the classification part, we have been used different classifications MNB, BNB, SGD, LR, SVC, Linear SVC, and CNN.

### 3.6. Evaluation

It is the last step in the text classification part. In this work, we have been used four evaluation matrices to prove this method is working properly. Accuracy, precision, recall, and F-Measure have been used. We will explain it in the experimental part 4.3.

## 4. Experimental analysis

In this part, we are going to explain the distribution of data and the implementation tools for this work. Initially, the dataset has many classes as expressed in Table 1.

### 4.1. Dataset

We have been used a dataset called Khalij, which balances as summarized in Table 1. The dataset is online available at this link*. The author for this dataset mention all details in [30,31]

### 4.2. Implementation environment

The proposed model implemented various experiments, the computation device is PC with the following specifications: Intel R © Core (TM) i7-6850 K processor with 24 GB RAM, 3.360 GHz frequency, Nvidia 1050Ti GPU. The algorithms such as MNB, BNB, SGD, LR, SVC, Linear SVC, and CNN were used for experimentation using Anaconda with Python 3.8.0. ML libraries like scikit-learn and NLTK have been used for investigation of the performance. In the subsequent sections we discuss on results. The codes are open sourced at GitHub†

* https://data.mendeley.com/datasets/57zpx667y9/1
† https://github.com/HPC

**Table 2**
Result using BOW in term of accuracy.

| Classification algorithm | Accuracy |
|---|---|
| Multinomial NB | 0.95 |
| Bernoulli NB | 0.90 |
| Logistic regression | 0.97 |
| SGD Classifier | 0.97 |
| SVC | 0.95 |
| Linear SVC | 0.97 |

**Table 3**
Result using TFIDF in terms of accuracy.

| Classification algorithm | Accuracy |
|---|---|
| Multinomial NB | 95 |
| Bernoulli NB | 90 |
| Logistic Regression | 97 |
| SGD Classifier | 97 |
| SVC | 97 |
| Linear SVC | 97 |

**Table 4**
Result using CNN with character level in terms of accuracy.

| Classification algorithm | Accuracy |
|---|---|
| Deep learning (CNN) | 0.98 |

### 4.3. Evaluation metrics

Where TP, TN, FP, and FN are known for number of true positive, true negative, false positive, and false negative detections, respectively. For obtaining above discussed parameters, a multidimensional confusion matrix utilized as in[22,32–34].

$$Recall = \frac{TP}{TP + FN} \tag{1}$$

$$Precision = \frac{TN}{TN + FP}, \tag{2}$$

$$F1 - score = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{3}$$

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \tag{4}$$

## 5. Results

Different Ml algorithms have been used with different feature extraction techniques for this experimentation. We will discuss here the experimental results as we have seen in Tables 2–4.

### 5.1. Representation by BOW

The classification results were noted in Table 2. In terms of accuracy metrics with different classification algorithms. The Logistic Regression, SGD Classifier, and Linear SVC were the best comparing other models when we use BoW as feature engineering.

### 5.2. Representation by TF-IDF

Results for the classification task using TFIDF are presented in Table 3. The results have shown good performance. We did all experimentation with different metrics such as precision, recall, and accuracy but we discuss only accuracy for different models

**Table 5**
Comparison evaluation results with the latest works.

| Ref. | Accuracy |
|---|---|
| [31] | 96 |
| [35] | 97 |
| [24] | 97 |
| Proposed work | 98 |

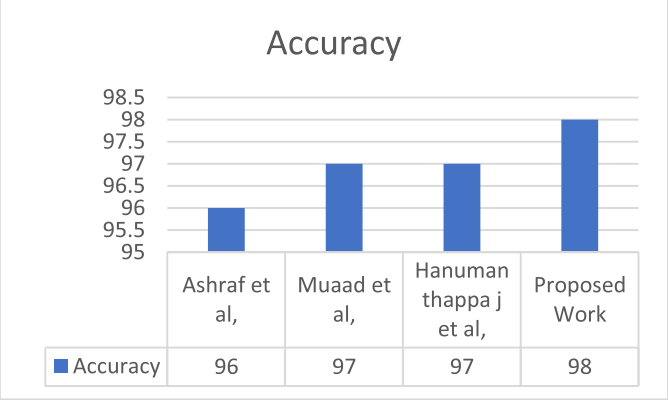



**Fig. 2.** Performance comparison of the proposed model.

### 5.3. Representation by character level

The classification result using Character-level representation model has been presented in Table 4. Here we can observe the best results when we compare it with other representation models.

## 6. Discussions

The evaluation of the performance of this work has been done using different feature extraction techniques with different representation models. We will explain this in this part. At the same time comparing with existing work which has been done for this dataset will be explored too.

Results using BOW, TF-IDF, and Character levels were explored in Tables 2–4. We compare and observed all of them and compare the result with three existing works as we see in Table 5 and Fig. 2.

We conclude that using character-level representation with CNN gives a good result that outperforms all others. With all advantage here and based on our research the need for a huge dataset and losing semantic meaning are still big challenges we plan to consider for future work.

## 7. Conclusion

In this paper, Arabic text classification has been proposed using the Khaleej dataset. Six machine learning algorithms and one deep learning algorithm has been employed to handle the task of Arabic text classification task. Different pre-processing methods have been done such as removing the stop word, and normalization after that stemming approach has been done too. The experimentation shows that the representation models and suitable feature engineering techniques enhanced the accuracy of Arabic text classification. The results have been proved that CNN outperforms all other models. There are different ideas for future work such as collecting more data, and working with different transfer learning models at the same time data augmentation for Arabic is still a big challenge to solve imbalance data. Finally increasing the accuracy of text classification using a hybrid ensembling technique is one more interesting area for Arabic text classification.

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